Calculating Efficiency

Efficiency is the term used to describe how well something does it's job and how little energy it requires to do that job. *For example:* the old style light bulb was not very efficient because most of the electrical energy (input) is wasted by the bulb (transducer) as heat (output).





A newer 'energy saving' bulb uses 18W of power to create the equivalent light of a 100W bulb.

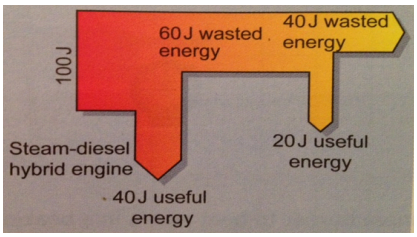
1) Calculate the energy used in Joules (J) for each of these bulbs. Each bulb is used for **2 hours:** *what is the* ***difference*** *in energy usage?*

Remember:
Energy (J) = power (W) x time (in seconds).

Knowing how efficient something is helps people to decide which device will cost less to use. If you were going shopping for some consumer electrical goods which one would you buy? Calculate their efficiency using the following formula.

$$\text{Efficiency} = \frac{\text{Useful energy out (J)}}{\text{Total energy in (J)}}$$

Device	Energy transfer	Efficiency (%)
	Input: 400J *Output* Sound: 225J Heat: 75J Light: 100J	Useful energy = Sound (225J) + Light (100J) 225+100=325J. Total output = 400 Efficiency = 325/400 = 0.8125 (x100) = 81%
	Input: 275J *Output* Light: 75J Sound: 90J Heat: 110J	
	Input: 1KJ *Output* Heat: 325J Sound: 300J Light: 375J	
	Input: 0.75KJ *Output* Heat: 400J Sound: 350J	
	Input: 2000J *Output* Heat: 1.5KJ Sound: 100J Light: 400J	





In a steam-diesel hybrid car, wasted energy is used to heat water and then produce steam; which is used as an input energy. The steam is used to power the car at slow speeds and conserves the diesel fuel.

1) Calculate the efficiency of this hybrid car using the data above.

2) Write an energy chain for the hybrid car *(input energy(s) – transducer – output energy(s)).*

3) Explain why this hybrid car is more efficient than a petrol only engine.

4) What could be done to a car to make it more efficient?

Keyword	***Meaning***	***Example***
Efficiency		
Energy		
	Something that changes an input energy into useful and not useful energies.	
		Petrol, coal and gas are examples of this type of energy.
Power		